

Submit your Jupyter notebook with the answers to gradescope by 1:50pm.

1. Let $p = 7x^8 - 4x^7 + 5x^6 - 7x^5 + 4x^4 - 6x^3 - 4x^2 + 3x + 1$.
 - (a) Construct the Horner form of p .
 - (b) Make a fast callable object of the Horner form of p .
 - (c) Compare the time it takes to evaluate p at 1.0 with the time to evaluate the fast callable of p at 1.0. Which form is faster to evaluate?
2. Consider the expression $q = \frac{x^3 + 3y^2 + 2}{y^3 + 3x^2 + 2}$. Draw the tree which represents q .